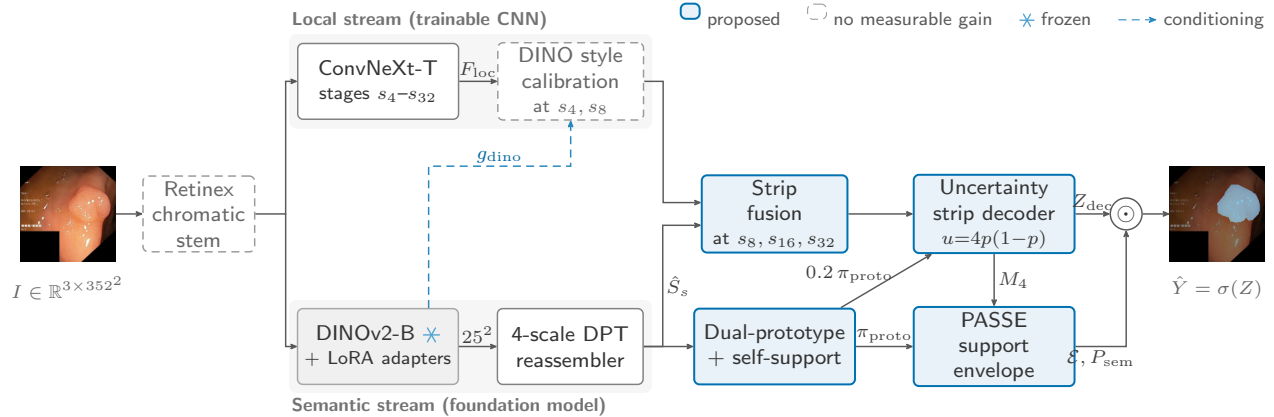

Local stream (trainable CNN)
proposed
no measurable gain
frozen
conditioning
ConvNeXt-T
stages s_4–s_{32}
F_{loc}
DINO style calibration
at s_4, s_8
g_{dino}
Retinex chromatic stem
$I \in \mathbb{R}^{3\times 352^2}$
Strip fusion
at s_8, s_{16}, s_{32}
Uncertainty strip decoder
$u{=}4p(1{-}p)$
Z_{dec}
$\hat{S}_s$
$0.2\,\pi_{\text{proto}}$
M_4
DINOv2-B
+ LoRA adapters
25^2
4-scale DPT reassembler
Dual-prototype + self-support
π_{proto}
PASSE support envelope
$\mathcal{E}$, P_{sem}
$\hat{Y} = \sigma(Z)$
Semantic stream (foundation model)